

Figure 1 -- The four IEEE benchmark systems, solved with full Newton-Raphson AC power flow.

Voltage spread is the first nonlinearity the DC model discards; graph diameter grows 5 -> 14 hops, which is what breaks message passing (Part 2).

case14 N = 14, diameter = 5 hops **case30** N = 30, diameter = 6 hops **case57** N = 57, diameter = 12 hops **case118** N = 118, diameter = 14 hops

